

Reduce production or improve outflow

Drug class helps predict both the pressure effect and the adverse effects to ask about.

Mechanism map

Class	Main action / recognisable clue
Beta blocker	Reduces aqueous production; systemic beta-blocking effects
Carbonic anhydrase inhibitor	Reduces secretion by inhibiting the production pathway
Prostaglandin analogue	Increases uveoscleral outflow; iris darkening and lash growth
Alpha-2 agonist	Reduces production and increases uveoscleral outflow

Match class to the patient

Different mechanisms can be combined when needed. Miotics such as pilocarpine constrict the pupil and may be used during angle-closure treatment; they can cause myopia, visual disturbance and headache. A pressure response alone does not prove that the medication is appropriate for continued use. Review tolerability, adherence and the systemic history, and assess nerve and field progression before deciding whether the overall treatment goal has been achieved.

Systemic history changes prescribing

Topical delivery does not remove the risk of clinically important systemic adverse effects.

High-yield safety checks

Timolol

Avoid with asthma, significant bradycardia or heart block; beta blockade can cause bronchoconstriction and cardiac effects.

Brimonidine

Avoid in young children because central effects may include respiratory depression.

Systemic acetazolamide

Review renal function and adverse effects, including paraesthesia, bicarbonate loss and metabolic acidosis.

Use adverse-effect patterns

Iris pigmentation and longer eyelashes point towards prostaglandin therapy. New wheeze or a slow pulse warrants review of topical beta blockade. Acetazolamide inhibits renal carbonic anhydrase as well as aqueous production, explaining bicarbonate loss and acidosis. Ask about all glaucoma and systemic medicines rather than focusing only on the most recent pressure number. Urgent glaucoma treatment still requires attention to contraindications when selecting agents.